

Report: Optimizing NYC Taxi Operations

Include your visualisations, analysis, results, insights, and outcomes. Explain your methodology and approach to the tasks. Add your conclusions to the sections.

1. Data Preparation

1.1. Loading the dataset

1.1.1. Sample the data and combine the files

Since the actual records per parquet file was insanely high, I choose to sample the records based on picking up hourly data from each day each month. Initial sampling fraction was chosen as 0.05.

After sampling record count was found to be around 18 lakh records.

To further scale down record size, out of this I choose to reduce the size to around 270000 so with this as my target size I further sampled the data so that my record count will be 270000.

This sampled data was stored in csv file ('final_sampled.csv') and df_new was the data frame I used for further analysis.

2. Data Cleaning

2.1. Fixing Columns

2.1.1. Fix the index

Column names were cleaned by stripping spaces and ensuring consistent formatting.

2.1.2. Combine the two airport_fee columns

The dataset contained two columns : 'airport_fee' and 'Airport_fee' which represented identical data for the records.

To combine both, I choose to create a new column : 'airport_fee_combined' and out of the two I choose max and updated the same in combined column.

After this old 'airport_fee' and 'Airport_fee' were dropped.

No records found where fare_amount < 0

2.2. Handling Missing Values

2.2.1. Find the proportion of missing values in each column

	missing_count	missing_percentage
congestion_surcharge	9412	3.410145
airport_fee_combined	9400	3.405797
passenger_count	9398	3.405072
RatecodeID	9398	3.405072
store_and_fwd_flag	9398	3.405072
improvement_surcharge	15	0.005435
mta_tax	14	0.005072

2.2.2. Handling missing values in passenger_count

I choose to impute null values in passenger_count with most frequently occurring which was 1. This approach maintains the distribution without skewing the data.

Also, I found out about 4298 rows were having 0 passenger_count but fare_amount > 0. This I felt is a data quality issue.

So I have proposed to change passenger_count to 1 as 1 being mode (most occurring) for records where fare_amount > 0

And rest where passenger_count is 0 and fare_amount is 0 were invalid so removed.

2.2.3. Handle missing values in RatecodeID

As ratecodeid is categorical I have chosen to keep NaN values as some category with -1 as code

2.2.4. Impute NaN in congestion_surcharge

Similarly for congestion_surcharge, I chosen to use mode value as impute for Nan values, as it keeps distribution intact.

This approach I feel is suitable for categorical data as it preserves the most common pattern of the dataset without any bias.

2.3. Handling Outliers and Standardising Values

2.3.1. Check outliers in payment type, trip distance and tip amount columns

I used IQR method to analyse outlier data and found these many outliers

[323]:	outlier_count
passenger_count	61896
payment_type	58953
trip_distance	36251
total_amount	31425
fare_amount	28510
RatecodeID	24339
airport_fee_combined	23427
tolls_amount	22258
tip_amount	21048
congestion_surcharge	20482
extra	5023
mta_tax	2632
improvement_surcharge	366
VendorID	68
PULocationID	0
DOLocationID	0

Payment_type:

Outliers for payment_type suggested are with 0 as they don't represent any categorical code. These were dropped from the dataset.

Trip_distance:

Outliers for trip_distance include either very short trips or extremely long ones.

Trips with distance < 0.1 miles but fare > \$300 were removed.

Trips with distance > 250 miles were also removed as extreme outliers.

Trips with 0 distance and fare, yet with different pickup and drop-off locations, were treated as invalid and removed.

3. Exploratory Data Analysis

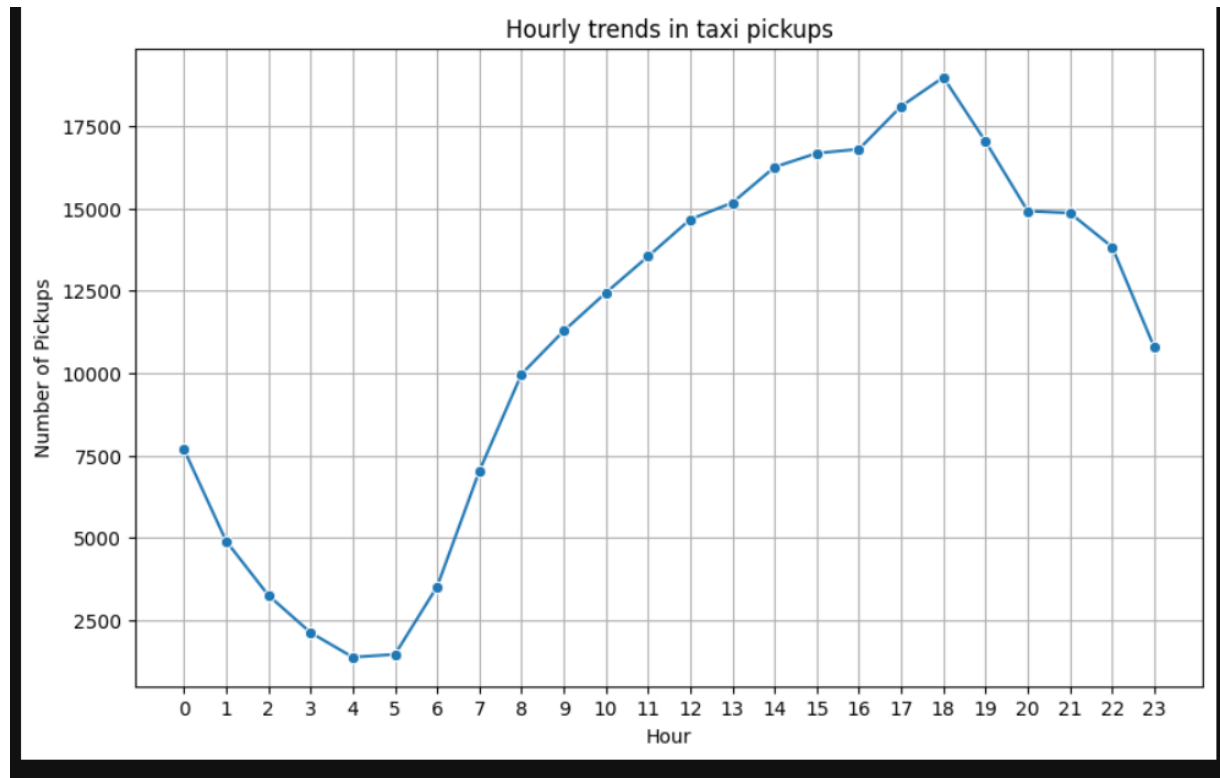
3.1. General EDA: Finding Patterns and Trends

3.1.1. Classify variables into categorical and numerical

Variable Name	Type
VendorID	Categorical
tpep_pickup_datetime	Datetime
tpep_dropoff_datetime	Datetime
passenger_count	Numerical
trip_distance	Numerical
RateCodeID	Categorical
PULocationID	Categorical
DOLocationID	Categorical
payment_type	Categorical
pickup_hour	Categorical
trip_duration	Numerical
fare_amount	Numerical
extra	Numerical
mta_tax	Numerical
tip_amount	Numerical
tolls_amount	Numerical
improvement_surcharge	Numerical
congestion_surcharge	Numerical
airport_fee	Numerical
total_amount	Numerical

3.1.2. Analyse the distribution of taxi pickups by hours, days of the week, and months

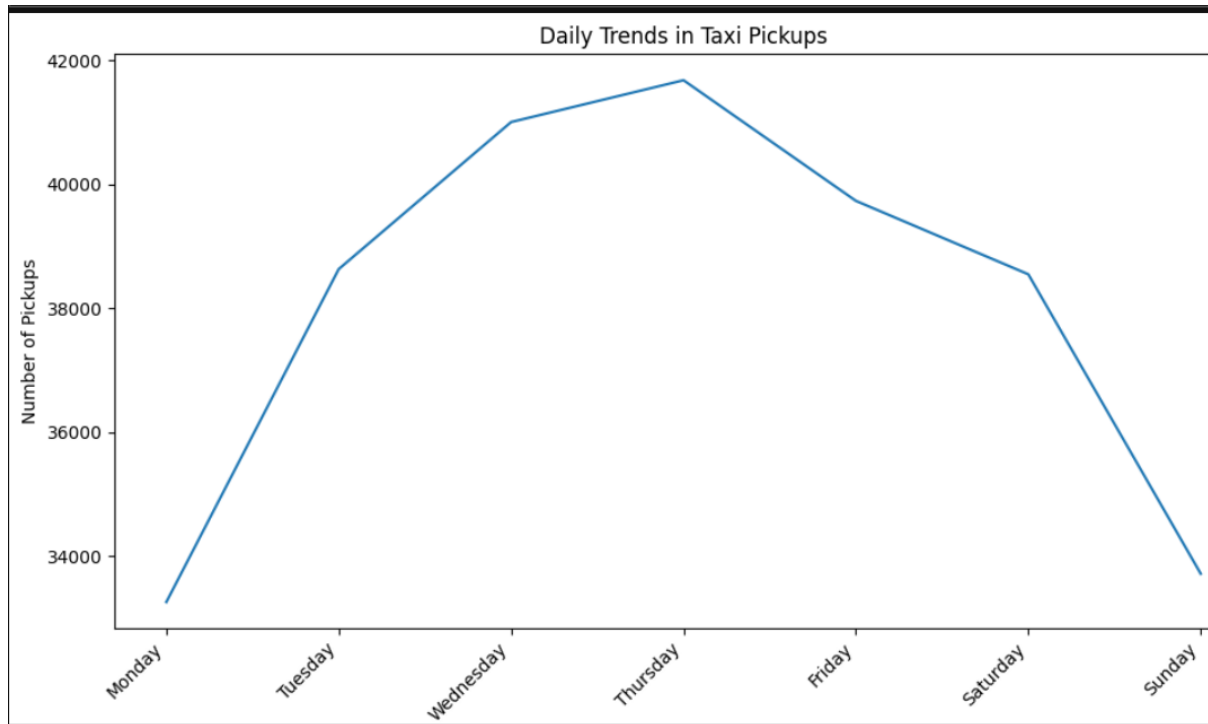
3.1.2.1. Taxi Pickup by Hour



Observations:

1. **18:00** busiest hour having around 18k pickups indicating peak demand mostly because of commute
2. Morning 4 and 5 relaxed hour indicating less demand during early morning hours

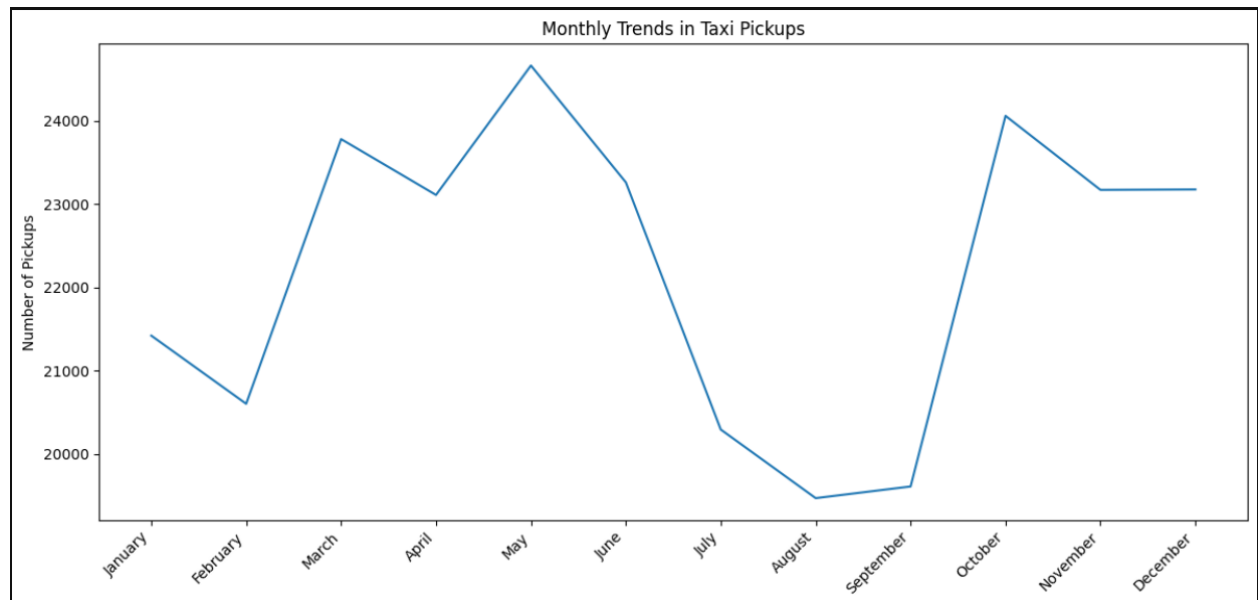
3.1.2.2 Taxi Pickup by Days of week



Observations:

1. **Thursday** is busiest day for pickups
2. Sunday Monday sees lowest demand

3.1.2.3 Taxi Pickup trend by month



Observations: **May is busiest month showing peak demand**

3.1.3. Filter out the zero/negative values in fares, distance and tips

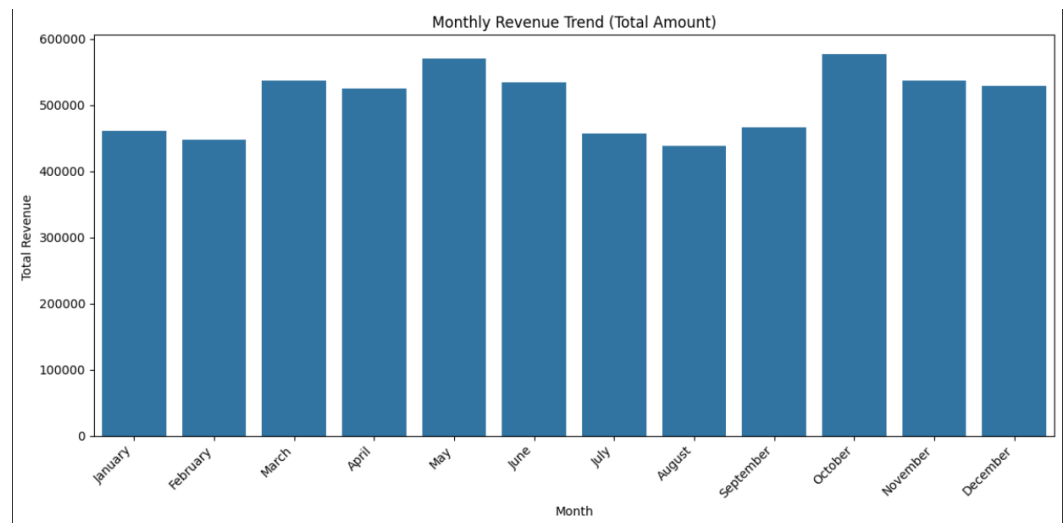
For $\text{total_amount} < 0$, might indicate a case of refund to be paid back in case of over payment or false payments, so I proposed to keep such transactions as boolean flag to handle.

For $\text{fare_amount} < 0$, it indicates invalid data so imputed with 0.

However, I retained zero tip_amount values, since tipping is optional and a large number of valid trips had no tip recorded. Many such entries still had a valid total amount, confirming they were legitimate.

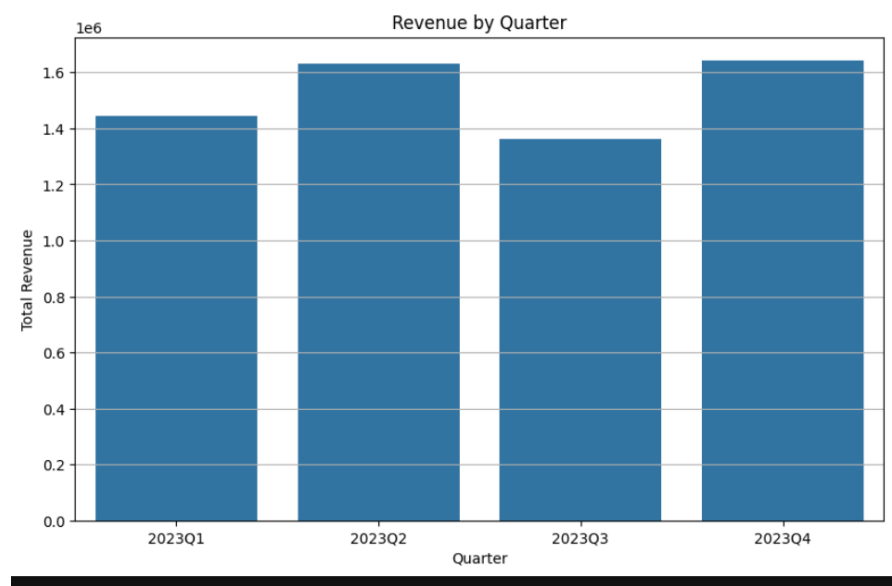
For trip_distance , entries where the value was zero despite different pickup and dropoff locations were considered inconsistent and were removed.

3.1.4. Analyse the monthly revenue trends



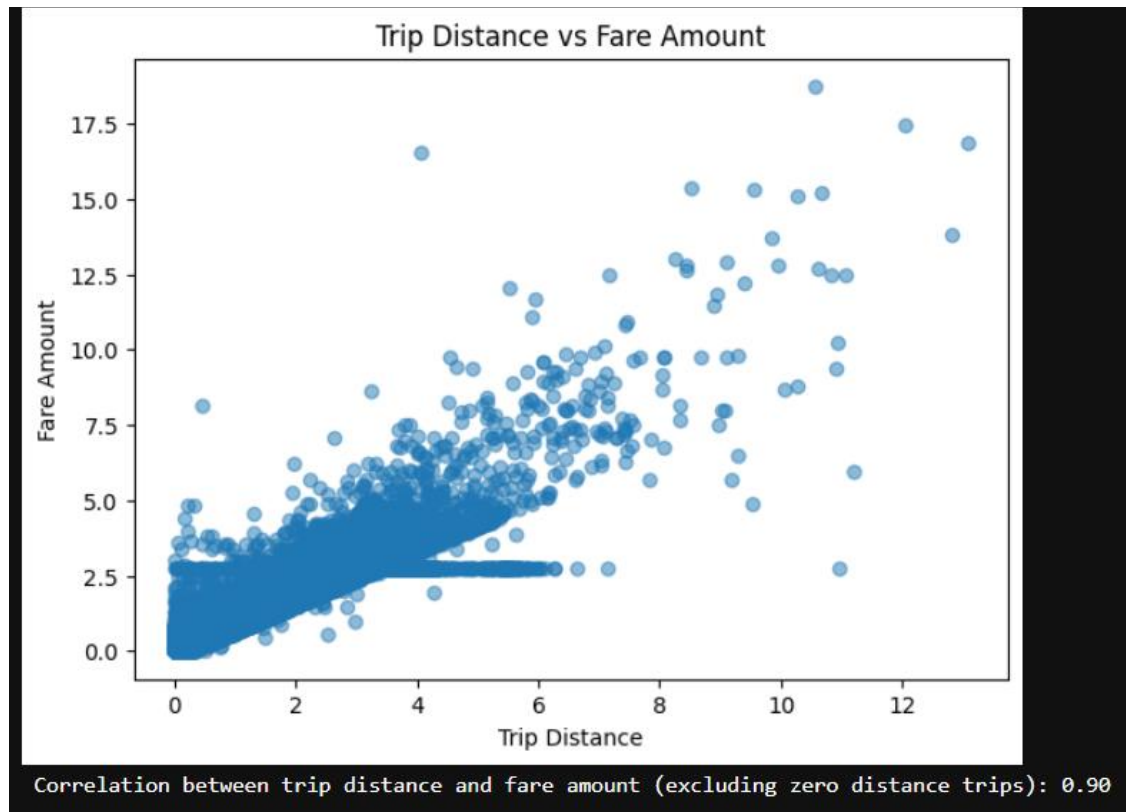
Observation: Highest revenue collected in October

3.1.5. Find the proportion of each quarter's revenue in the yearly revenue



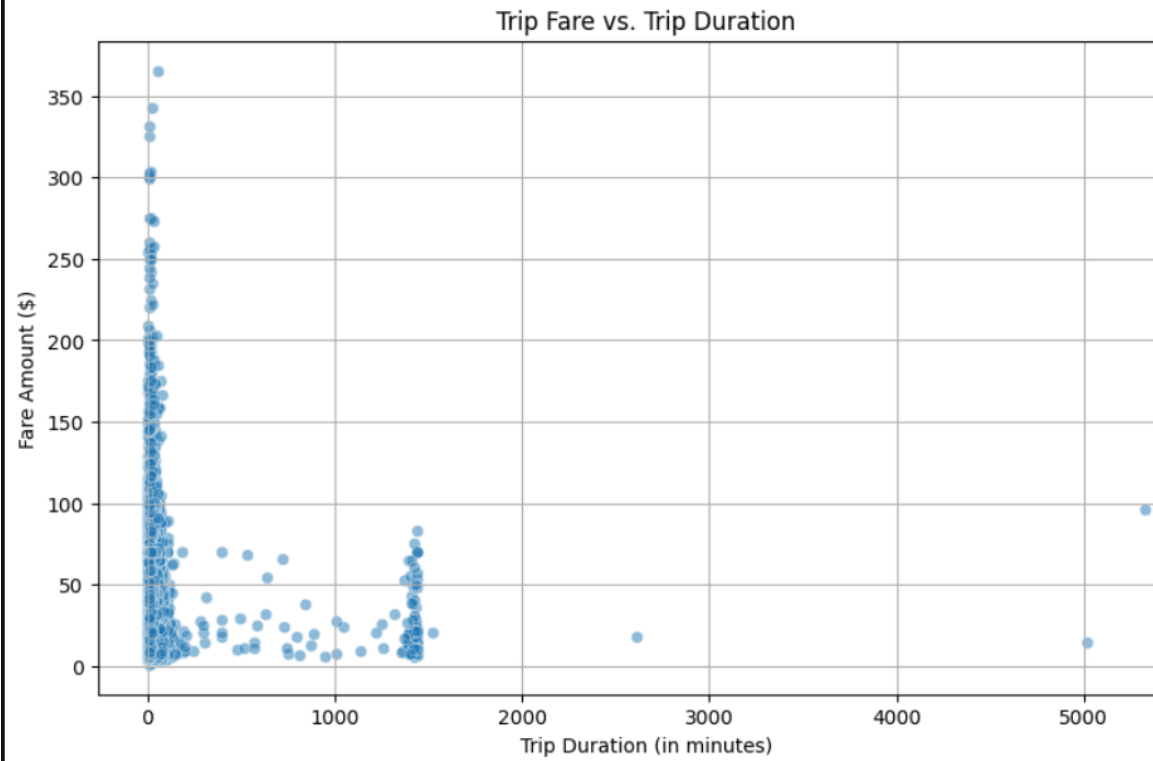
Observation: **Q4 wins in terms of revenue**

3.1.6. Analyse and visualise the relationship between distance and fare amount

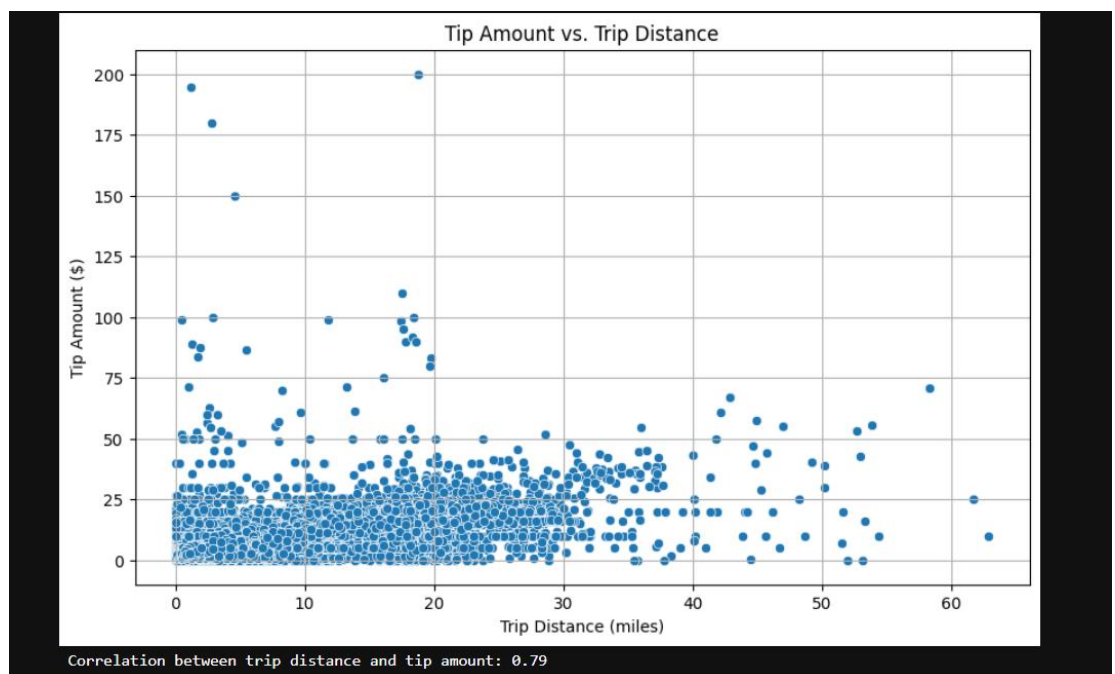
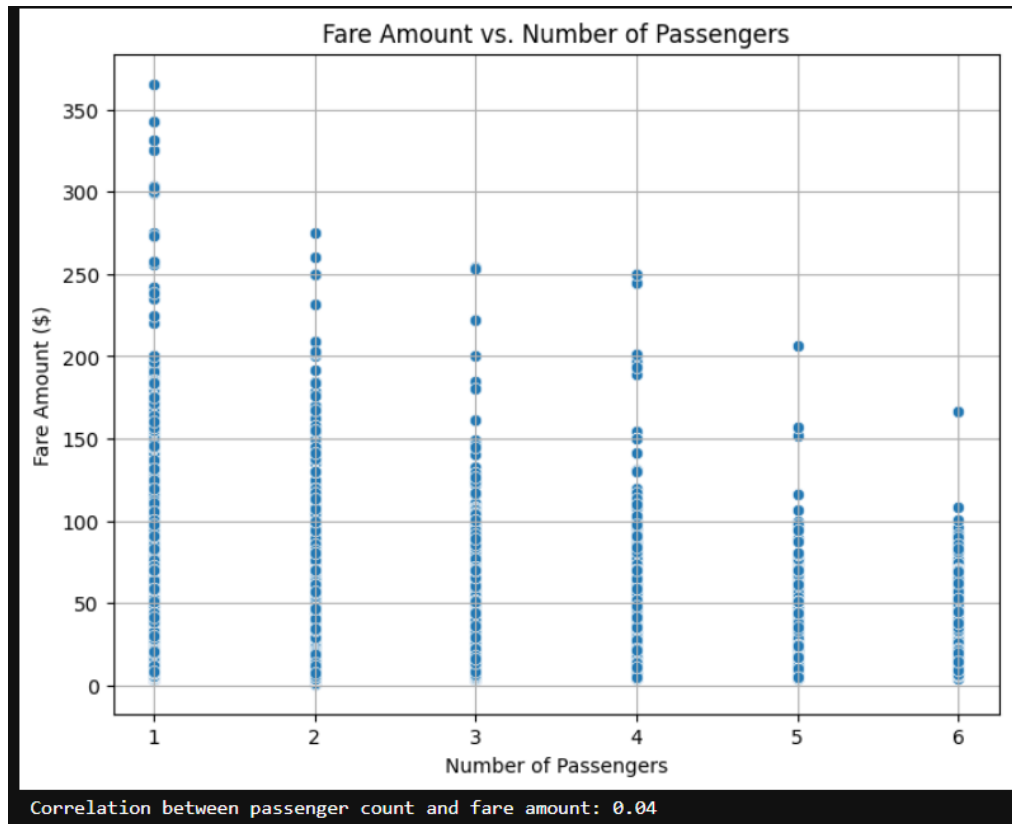


Observation: Strong positive correlation between trip distance and fare amount (0.90) indicates This confirms that **fare pricing is largely distance-driven**, as expected.

3.1.7. Analyse the relationship between fare/tips and trips/passengers

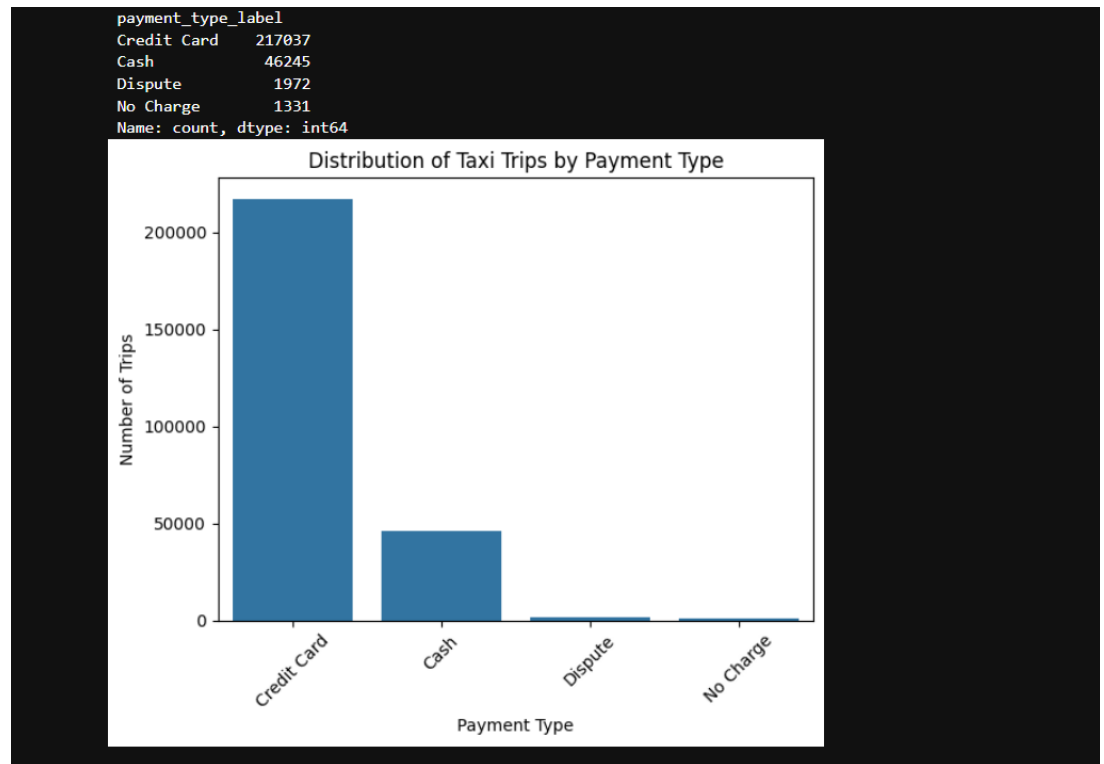


Correlation between trip duration and fare amount: 0.004470954566379133



Observation: Strong positive correlation between tip Amount and trip distance indicates larger trips generally implies good tip from customer.

3.1.8. Analyse the distribution of different payment types

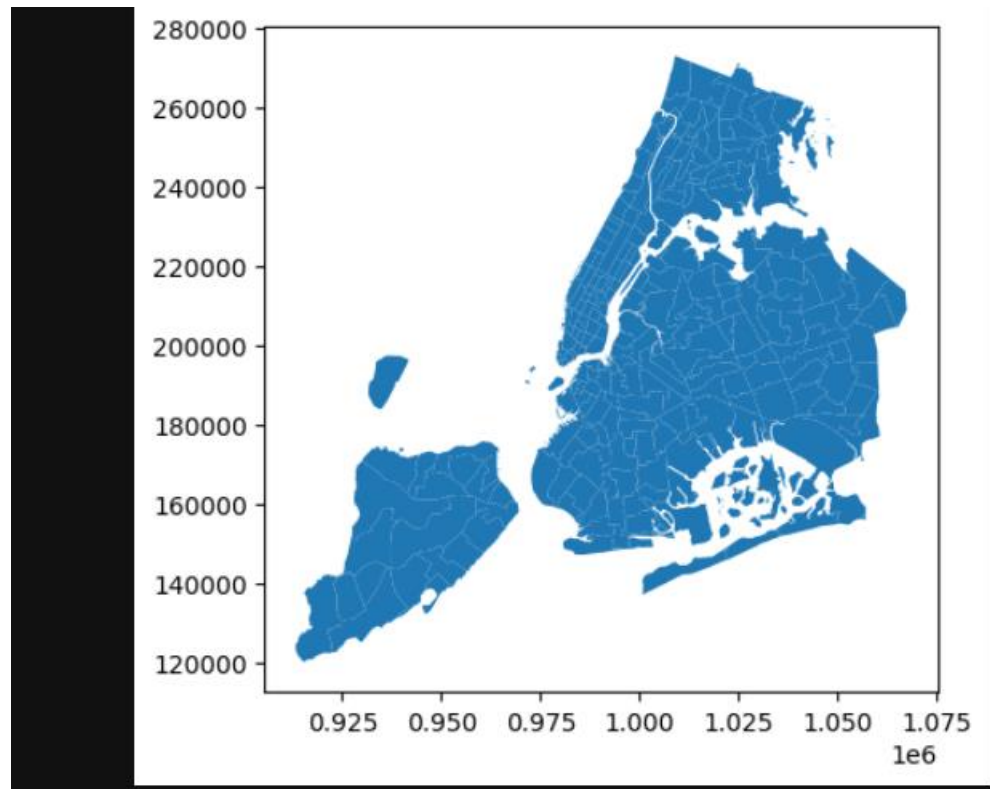


Observation: Payment mode is dominated by Credit card.

Can be a great option to introduce loyalty points to encourage digital payments and providing customer with cashback.

3.1.9. Load the taxi zones shapefile and display it

[368]:	OBJECTID	Shape_Leng	Shape_Area	zone	LocationID	borough	geometry
0	1	0.116357	0.000782	Newark Airport	1	EWB	POLYGON ((933100.918 192536.086, 933091.011 19...
1	2	0.433470	0.004866	Jamaica Bay	2	Queens	MULTIPOLYGON (((1033269.244 172126.008, 103343...
2	3	0.084341	0.000314	Allerton/Pelham Gardens	3	Bronx	POLYGON ((1026308.77 256767.698, 1026495.593 2...
3	4	0.043567	0.000112	Alphabet City	4	Manhattan	POLYGON ((992073.467 203714.076, 992068.667 20...
4	5	0.092146	0.000498	Arden Heights	5	Staten Island	POLYGON ((935843.31 144283.336, 936046.565 144...



3.1.10. Merge the zone data with trips data

Left join on LocationID column from zone data and trips data.

3.1.11. Find the number of trips for each zone/location ID

[373]:

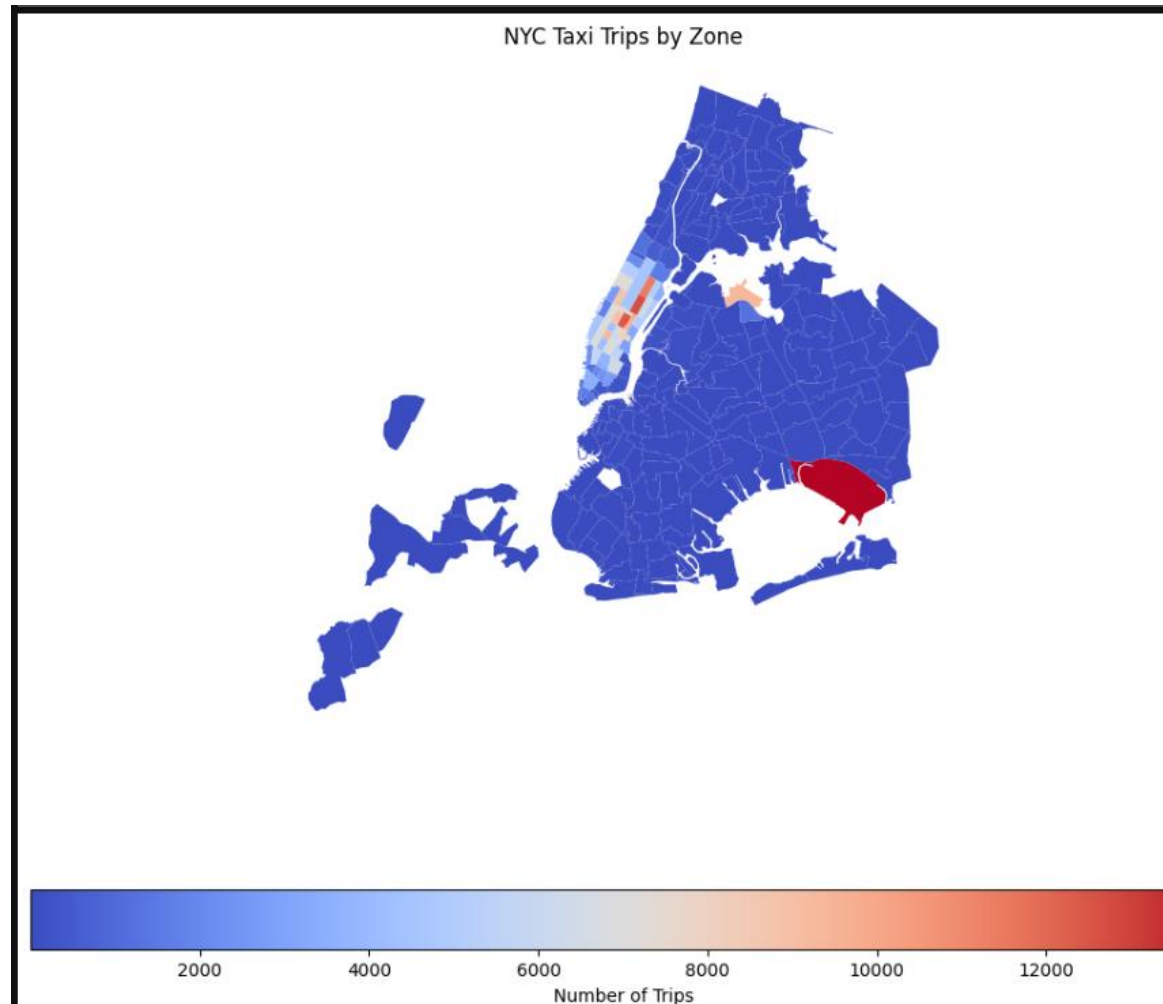
	PULocationID	num_trips
0	1	29
1	3	9
2	4	314
3	5	4
4	6	4
...	...	...
235	261	1447
236	262	3420
237	263	5092
238	264	2590
239	265	162

240 rows × 2 columns

3.1.12. Add the number of trips for each zone to the zones dataframe

OBJECTID	Shape_Leng	Shape_Area	zone	LocationID	borough	geometry	PULocationID	num_trips	
0	1	0.116357	0.000782	Newark Airport	1	EWB	POLYGON ((933100.918 192536.086, 933091.011 19...	1.0	29.0
1	2	0.433470	0.004866	Jamaica Bay	2	Queens	MULTIPOLYGON (((1033269.244 172126.008, 103343...	NaN	NaN
2	3	0.084341	0.000314	Allerton/Pelham Gardens	3	Bronx	POLYGON ((1026308.77 256767.698, 1026495.593 2...	3.0	9.0
3	4	0.043567	0.000112	Alphabet City	4	Manhattan	POLYGON ((992073.467 203714.076, 992068.667 20...	4.0	314.0
4	5	0.092146	0.000498	Arden Heights	5	Staten Island	POLYGON ((935843.31 144283.336, 936046.565 144...	5.0	4.0

3.1.13. Plot a map of the zones showing number of trips



3.1.14. Conclude with results

- Distance and fare show strong correlation (**approx 0.9**) confirming that fare is priced as per distance
- **Evening 6** indicates busiest hour of the day.
- **Thursday** shows peak demand during the whole week, indicating occurrence of special event
- **May** shows busiest month in terms of rides
- Quarter wise **2023Q4** shows highest revenue
- Airport and Midtown zones have highest pickup/drop-off for the rides, indicating busiest zone
- Most trips have 1 or 2 as frequent travellers and Credit card dominates payment mode.
- Jamaica Bay shows no rides indicating less business in that zone.

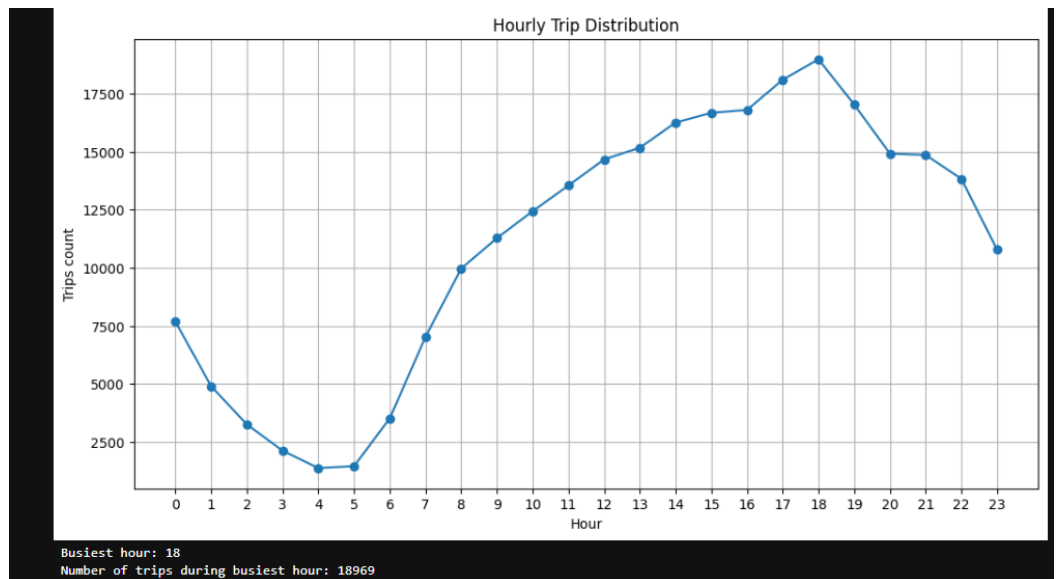
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3.2. Detailed EDA: Insights and Strategies

3.2.1. Identify slow routes by comparing average speeds on different routes

	PULocationID	DOLocationID	pickup_hour	avg_speed_mph
42779	194	194	15	0.002741
60744	264	132	14	0.008469
28461	142	142	5	0.023770
32085	151	24	1	0.032131
16759	113	235	22	0.048105
42780	194	194	16	0.048913
31843	148	232	12	0.051573
4682	48	229	0	0.061705
22515	137	68	20	0.071780
6158	67	215	15	0.078091

3.2.2. Calculate the hourly number of trips and identify the busy hours



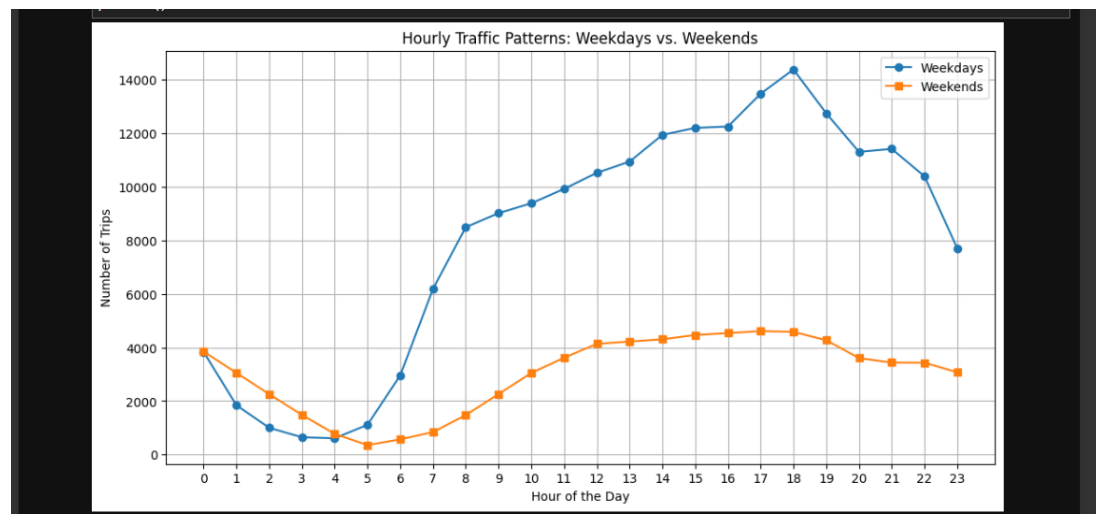
3.2.3. Scale up the number of trips from above to find the actual number of trips

I sampled data at 0.05 for first stage and then took target size = 270000 and estimated second sampling fraction

Thus effective sampling fraction : $0.05 * (\text{target_size} / \text{current_size})$

```
0.007276945792026999
[379]: pickup_hour
18    133938
17    127867
19    119662
16    117512
15    117407
Name: count, dtype: int64
```

3.2.4. Compare hourly traffic on weekdays and weekends



Observations:

- Weekdays traffic mostly rises from 6 to 9 AM drastically, which indicates heavy demand due to morning office commute, similarly 5 to 7 PM in evening indicate return commute.
- Weekends are more leisure based, people move out for entertainment, night life hence peak between 3 to 6 PM
- Fleet density can be reduced during late night on weekdays, but sufficient coverage to be maintained on weekends

3.2.5. Identify the top 10 zones with high hourly pickups and drops

Top 10 Pickup Zones:			
[382]:	LocationID	Pickup_Trips	zone
0	132	14016	JFK Airport
1	237	12694	Upper East Side South
2	161	12416	Midtown Center
3	236	11427	Upper East Side North
4	162	9571	Midtown East
5	138	9359	LaGuardia Airport
6	186	9294	Penn Station/Madison Sq West
7	230	8902	Times Sq/Theatre District
8	142	8791	Lincoln Square East
9	170	7898	Murray Hill

Top 10 Dropoff Zones:			
[383]:	LocationID	Dropoff_Trips	zone
0	236	11837	Upper East Side North
1	237	11132	Upper East Side South
2	161	10403	Midtown Center
3	230	8345	Times Sq/Theatre District
4	170	7803	Murray Hill
5	162	7642	Midtown East
6	142	7583	Lincoln Square East
7	239	7318	Upper West Side South
8	141	7063	Lenox Hill West
9	68	6695	East Chelsea

3.2.6. Find the ratio of pickups and dropoffs in each zone

```

top_pickup_dropoff_ratio

[385]: zone
      East Elmhurst      8.149225
      JFK Airport      4.263553
      LaGuardia Airport  2.620624
      Penn Station/Madison Sq West  1.556960
      Greenwich Village South  1.376644
      Central Park      1.368748
      West Village      1.332370
      Midtown East      1.242286
      Midtown Center    1.187999
      Garment District   1.186987
      Name: pickup_dropoff_ratio, dtype: float64

[239]: bottom_pickup_dropoff_ratio

[239]: zone
      Freshkills Park      0.000000
      Broad Channel      0.000000
      Oakwood      0.000000
      West Brighton      0.032258
      Newark Airport      0.037953
      Grymes Hill/Clifton  0.040000
      Breezy Point/Fort Tilden/Riis Beach  0.051282
      Windsor Terrace      0.053952
      Mariners Harbor      0.068966
      Whitestone      0.071038
      Name: pickup_dropoff_ratio, dtype: float64

```

3.2.7. Identify the top zones with high traffic during night hours

```

top_night_pickups

[388]: pickup_zone
      East Village      2210
      JFK Airport      1998
      West Village      1769
      Clinton East      1383
      Lower East Side    1355
      Greenwich Village South  1318
      Times Sq/Theatre District  1147
      Penn Station/Madison Sq West  898
      East Chelsea      869
      LaGuardia Airport  852
      Name: PULocationID, dtype: int64

[242]: top_night_dropoffs

[242]: dropoff_zone
      East Village      1148
      Clinton East      1006
      Murray Hill      881
      Gramercy      867
      Lenox Hill West    803
      East Chelsea      796
      West Village      740
      Yorkville West      691
      Lower East Side    667
      Sutton Place/Turtle Bay North  628
      Name: dolocationid, dtype: int64

```

3.2.8. Find the revenue share for nighttime and daytime hours

```
Nighttime Revenue Share: 12.11%  
Daytime Revenue Share: 87.89%
```

Pricing Strategy

3.2.9. For the different passenger counts, find the average fare per mile per passenger

```
[390]: passenger_count  
1.0    16.776528  
2.0    10.085446  
3.0     6.340590  
4.0     5.636363  
5.0     3.078471  
6.0     2.150627  
Name: fare_per_mile_per_passenger, dtype: float64
```

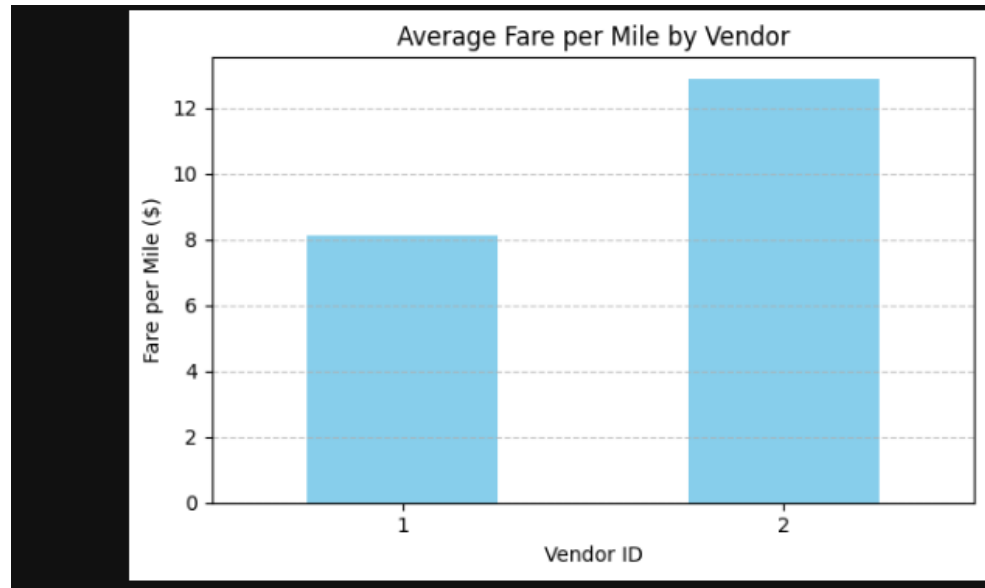
3.2.10. Find the average fare per mile by hours of the day and by days of the week

```
[391]: day_of_week  
Monday    16.97  
Tuesday    16.47  
Wednesday  17.71  
Thursday   17.71  
Friday     16.93  
Saturday   16.70  
Sunday     19.89  
Name: fare_per_mile, dtype: float64
```

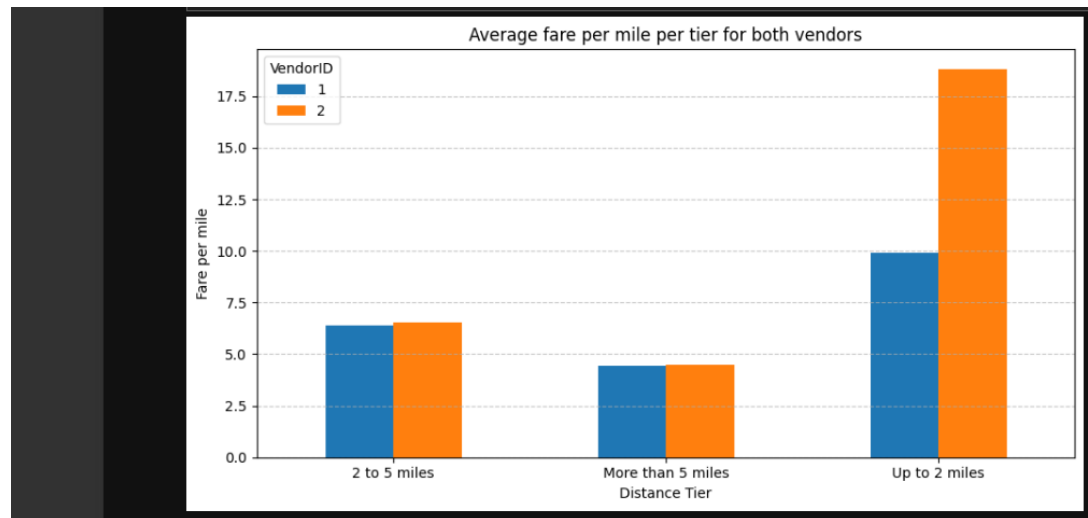
3.2.11 [3 marks]

Analyse the average fare per mile for the different vendors for different hours of the day

3.2.11. Analyse the average fare per mile for the different vendors



3.2.12. Compare the fare rates of different vendors in a distance-tiered fashion



3.2.13. Analyse the tip percentages

```

Average Tip Percentage by Distance:
distance_category
Up to 2 miles      12.127744
2 to 5 miles      12.313140
More than 5 miles  11.279590
Name: tip_percentage, dtype: float64

```

```

Average Tip Percentage by Passenger Count:
passenger_category
1 passenger      12.153279
2-3 passengers   11.695456
4+ passengers    11.346980
Name: tip_percentage, dtype: float64

```

```

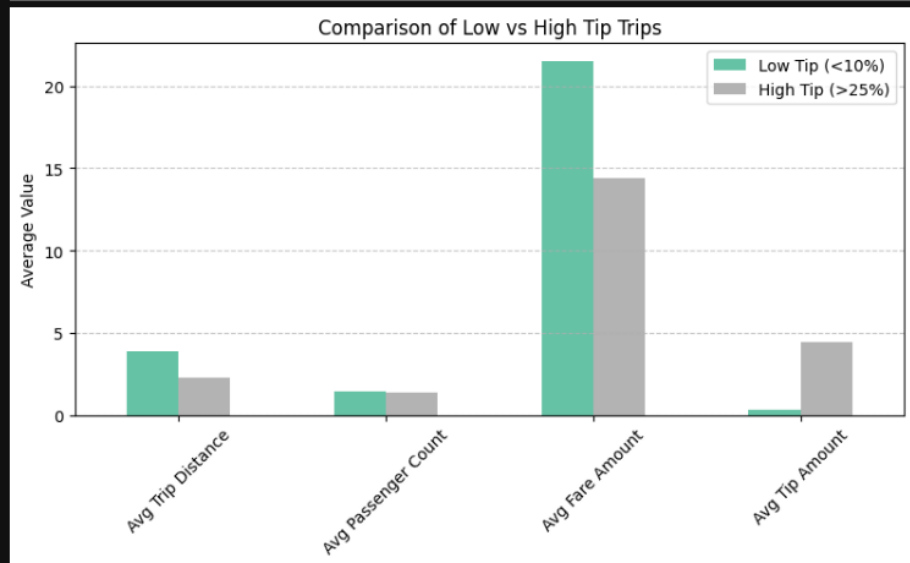
Average Tip Percentage by Time of Pickup:
time_category
Midnight to 6 AM    11.600103
6 AM to Noon        11.922043
Noon to 6 PM        11.842507
6 PM to Midnight    12.404715
Name: tip_percentage, dtype: float64

```

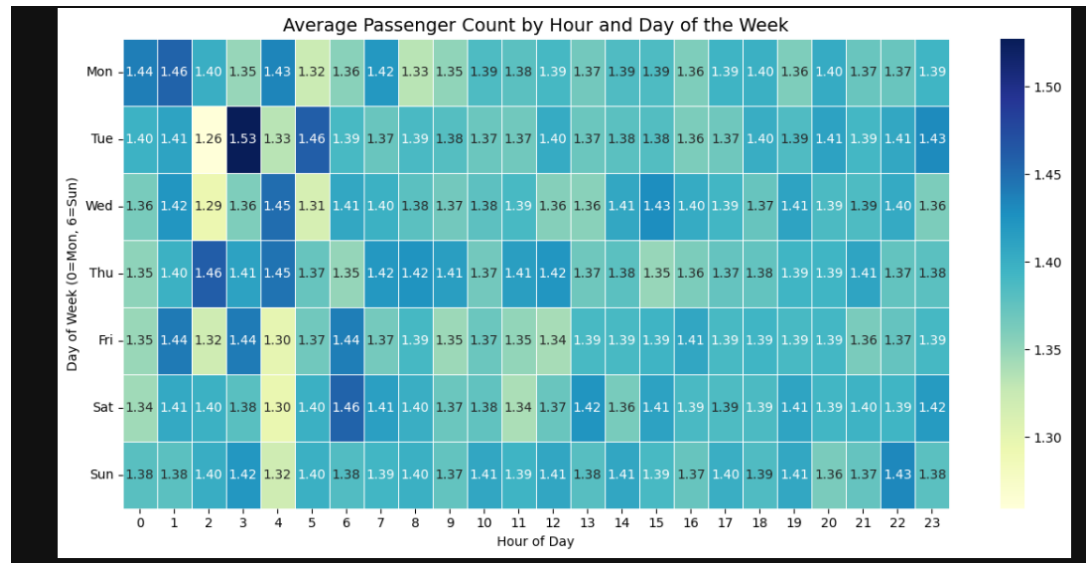
```

Most Common Low Tip Scenarios:
distance_category passenger_category time_category count
Up to 2 miles      1 passenger      Noon to 6 PM    14755
                  1 passenger      6 PM to Midnight 10723
                  1 passenger      6 AM to Noon    8951
2 to 5 miles      1 passenger      Noon to 6 PM    5960
                  1 passenger      6 PM to Midnight 5800
More than 5 miles 1 passenger      Noon to 6 PM    4030
2 to 5 miles      1 passenger      6 AM to Noon    4019
More than 5 miles 1 passenger      6 PM to Midnight 3814
Up to 2 miles      2-3 passengers  Noon to 6 PM    3683
                  2-3 passengers  6 PM to Midnight 3208

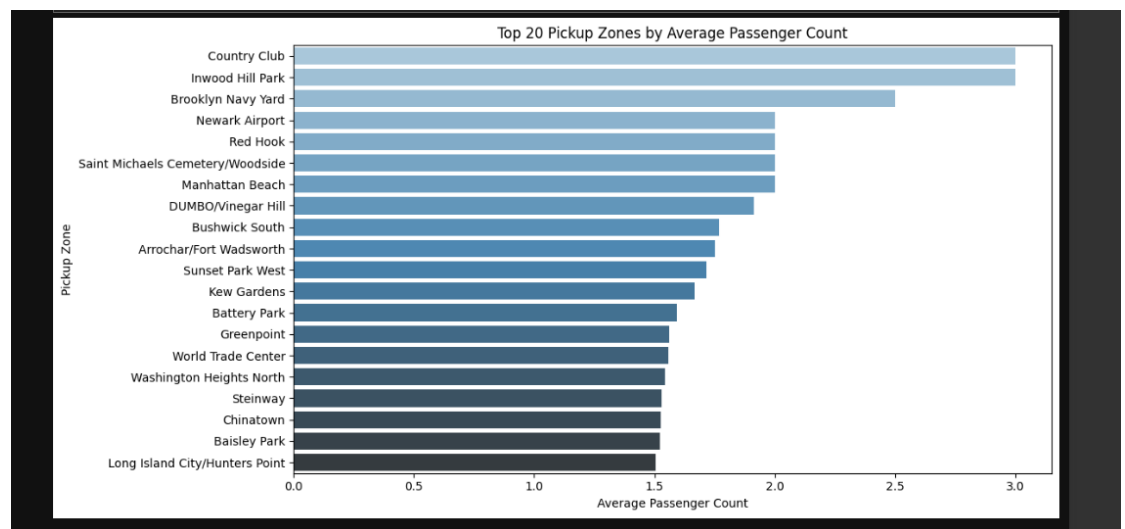
```



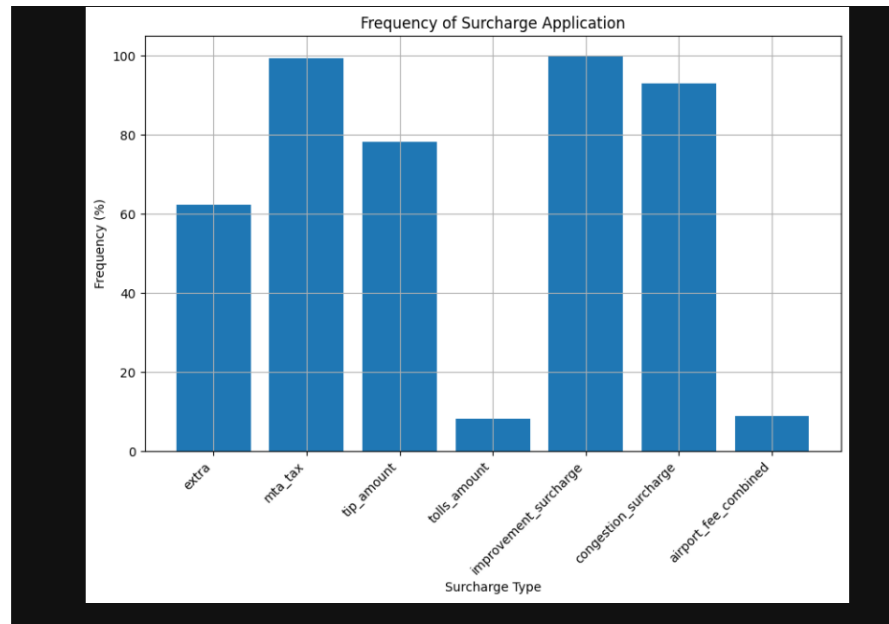
3.2.14. Analyse the trends in passenger count



3.2.15. Analyse the variation of passenger counts across zones



3.2.16. Analyse the pickup/dropoff zones or times when extra charges are applied more frequently.



4. Conclusions

4.1. Final Insights and Recommendations

4.1.1. Recommendations to optimize routing and dispatching based on demand patterns and operational inefficiencies.

- Recommendation:

Peak Hour strategy:

- Pre position cabs near high demand zones like Airports, Mid Town zones before peak hours.
- Reduce idle cruising during off-peak hours by consolidating dispatch zones
- Ensure sufficient supply of drivers and cabs during morning and evening commutes during weekday.

-Impact:

- Lower passenger waits times
- Reduced driver idle times
- Higher trip completion during peaks

Optimize dispatch of cabs as per days:

- Prioritize Business hubs, corporate parks during weekdays
- Prioritize Nightlife, entertainment zones.

-Impact:

- Faster pickups
- Lower wastage of fuel
- Improved driver earnings per hour

4.1.2. Suggestions on strategically positioning cabs across different zones to make best use of insights uncovered by analysing trip trends across time, days and months.

Time-based cab positioning:

- Increase cab availability during morning peak hours (office commute) and evening peak hours (return commute).
- Position more cabs near business districts, transport hubs, and residential exit zones during these times.
- Reduce fleet concentration in low-demand zones.

Day based strategy:

- Focus on office-centric zones, IT parks, commercial areas, and metro stations.
- Morning demand is stronger than weekends, so proactive positioning is critical.

Monthly / Seasonal Adjustments:

- During high-demand months (Festivals, Holidays) increase fleet size or position driver to longer working windows.
- We can divide zones based on entertainment, business, low-demand etc and use this to strategically place cabs targeting appropriate zones

:

4.1.3. Propose data-driven adjustments to the pricing strategy to maximize revenue while maintaining competitive rates with other vendors.

Based on the analysis of trip distance, fare amount, time-based demand patterns, and payment behavior, the following pricing strategy adjustments are recommended to maximize revenue while remaining competitive:

- Time based dynamic pricing with limited surge charge lower than competitors. Benefit we can obtain from this is increased revenue during high demand periods without compromising on price sensitizing riders.
- Since majority of trips are short trips (< 1 mile), we can propose reduced wait time incentive for drivers by assigning them multiple short trips and giving them incentives for reduced wait time
- As there is strong correlation between trip_distance and fare_amount, we should immediately flag anomalies and try to

reduce inconsistent pricing and should maintain transparent distance-based pricing.

- Payment type incentives for credit card holders. As credit card payments dominate significantly, we can introduce loyalty points and offer small cashbacks to encourage digital payment further which can improve operational efficiency and boost customer satisfaction.
- Certain zones and routes consistently generate high demand. So we can introduce zone based pricing premium for high pickup drop zones and keep pricing competitive in low-demand zones.